

CampusEats — Services, Contracts & Validation

1. Capabilities

1. **Authenticate Users** — verify users and provide secure access to CampusEats.
 2. **Browse Menus** — allow customers to view restaurants, menu items, prices, and availability.
 3. **Place Orders** — allow customers to select food items and submit orders.
 4. **Process Payments** — process customer payments and maintain payment status.
 5. **Manage Orders** — allow restaurant staff to accept, prepare, and complete orders.
 6. **Track Orders** — allow customers to view the current status of their orders.
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2. Service Contracts

2.1 Identity Service

Operation	Input	Output	Errors
<code>authenticateUser</code>	username, password	authentication result, user role	Invalid credentials, user not found

2.2 Catalogue Service

Operation	Input	Output	Errors
<code>listRestaurants</code>	campus/location	available restaurants	No restaurants available
<code>getMenu</code>	restaurant identifier	menu items, prices, availability	Restaurant not found
<code>checkItem</code>	restaurant identifier, item identifier, quantity	item availability, current price	Item not found, item unavailable

2.3 Order Service

Operation	Input	Output	Errors
<code>placeOrder</code>	user identifier, restaurant identifier, selected items, payment details	order confirmation, order status, payment status	Invalid item, item unavailable, payment failed
<code>getOrder</code>	order identifier	order details, current status	Order not found, unauthorized access
<code>updateOrderStatus</code>	order identifier, new status	updated order status	Order not found, invalid status transition

2.4 Payment Service

Operation	Input	Output	Errors
<code>authorizePayment</code>	order information, payment details, amount	payment result, payment status	Payment declined, invalid payment details

2.5 Notification Service

Operation	Input	Output	Errors
<code>sendOrderUpdate</code>	user identifier, order identifier, message	notification result	User not found, notification delivery failed

3. placeOrder — Full Specification

Purpose

`placeOrder` allows a customer to submit a food order and receive confirmation of the order and payment status.

Inputs

- User identifier
- Restaurant identifier
- Selected menu items and quantities
- Payment details

Outputs

- Order confirmation
- Order status
- Payment status

Error Cases

Error	Meaning
<code>UserNotFound</code>	The customer cannot be located or authenticated.
<code>RestaurantNotFound</code>	The selected restaurant is unavailable.
<code>ItemNotFound</code>	A requested menu item does not exist.
<code>ItemUnavailable</code>	A requested menu item is currently unavailable.
<code>PaymentFailed</code>	Payment authorization was unsuccessful.
<code>OrderCreationFailed</code>	The order could not be completed.

Internal Details Hidden From Callers

Callers do not need to know:

- how orders are stored;
- how menu items are stored;
- how item availability is checked internally;
- how payment processing is implemented;
- how order status is stored or changed;
- which database or technology is used;
- how services communicate internally.

Internal Flow

Customer calls `placeOrder`.

1. Order Service validates the requested items with Catalogue Service.
2. Order Service requests payment authorization from Payment Service.
3. Order Service creates the order after successful validation and payment.
4. Order Service returns the order confirmation and status.

4. Service Validation

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled	Fix
Identity Service	Yes	Yes	Yes	Yes	Yes	None
Catalogue Service	Yes	Yes	Yes	Yes	Yes	None
Order Service	Yes	Yes	Yes	Yes	No	Use service contracts for Catalogue and Payment communication; never share their internal data.
Payment Service	Yes	Yes	Yes	Yes	Yes	None
Notification Service	Yes	Yes	Yes	Yes	Yes	None